

Ethyl Acetate Synthesis via Reactive Distillation

Contribution	
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Using Considerable Excess of Acetic Acid — A Low-Energy Process

Group 2 | Chemical Process Technology

Aspen Plus Simulation & Design Optimization | April 2026

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Reactive Distillation

Aspen Plus

NRTL Model

Heat Integration

80% Energy Savings

Introduction & Motivation

⚠ THE PROBLEM

Global ETAC demand

2M+ tonnes/year

Conventional energy use

2.2–2.5 t steam / t ETAC

Key bottleneck

ETOH–ETAC–H₂O ternary azeotrope (70.5°C)

Separation challenge

Complex LLE + VLE, costly downstream

Catalyst issue

Conc. H₂SO₄ — corrosive, hazardous

✓ OUR APPROACH

Considerable excess HAC (15×)

Drives ETOH to near-complete conversion

No ETOH in overhead

Binary ETAC–H₂O azeotrope only → direct phase split

Molecular sieve membrane

Replaces energy-intensive extractive distillation

Heterogeneous solid acid catalyst (Amberlyst-36)

Eliminates H₂SO₄ corrosion problems

Energy target

1,149 kJ/kg → 80% reduction vs. conventional

Thermodynamic Framework — NRTL Model & Azeotrope Data

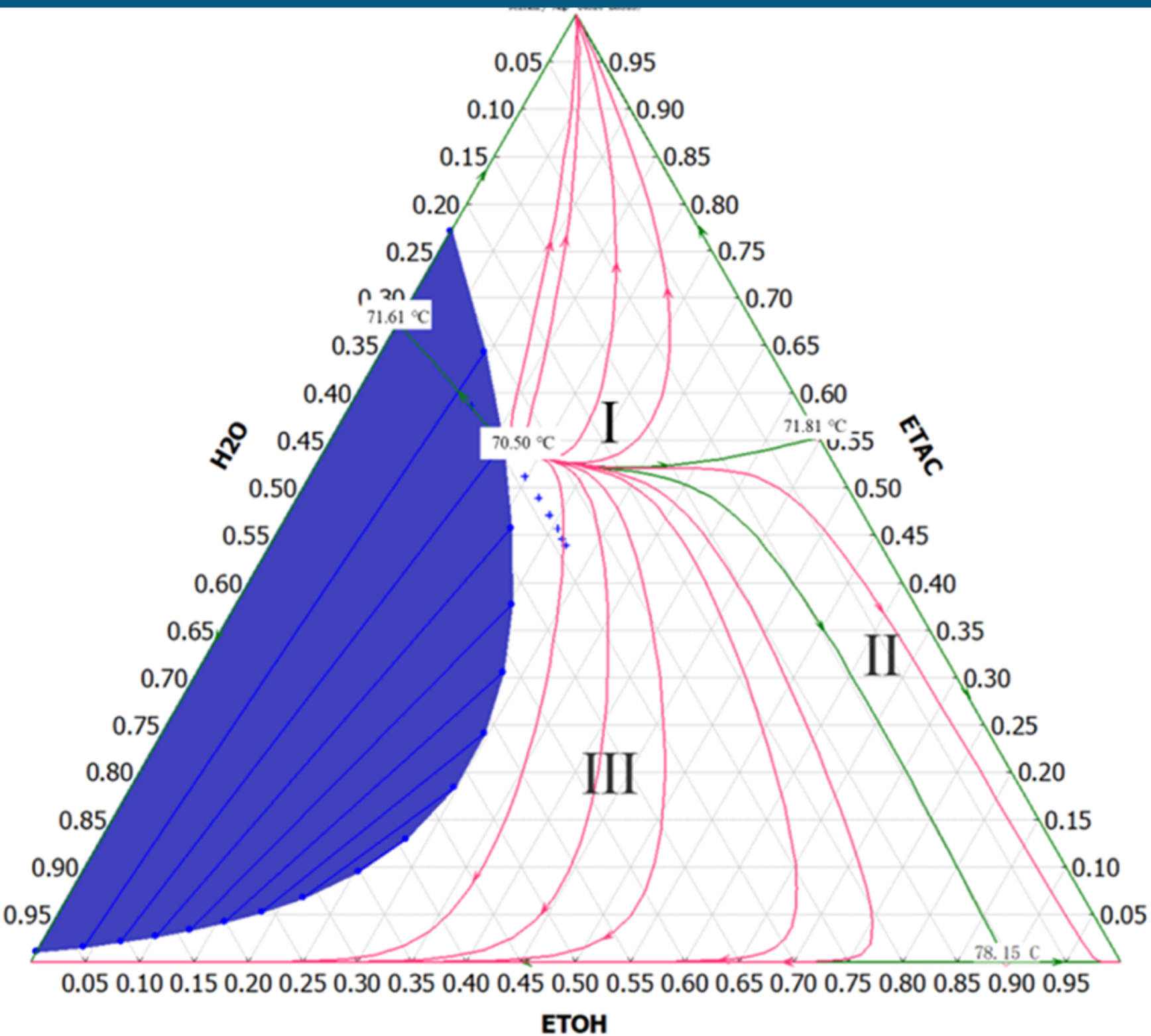
Property Method: NRTL

- Accounts for non-ideal VLE & LLE in quaternary system
- Binary params from: APV110 VLE-IG, VLE-HOC, LLE-ASPEN
- System operates at atmospheric pressure (1.01325 bar)

Azeotrope Data @ 101.325 kPa

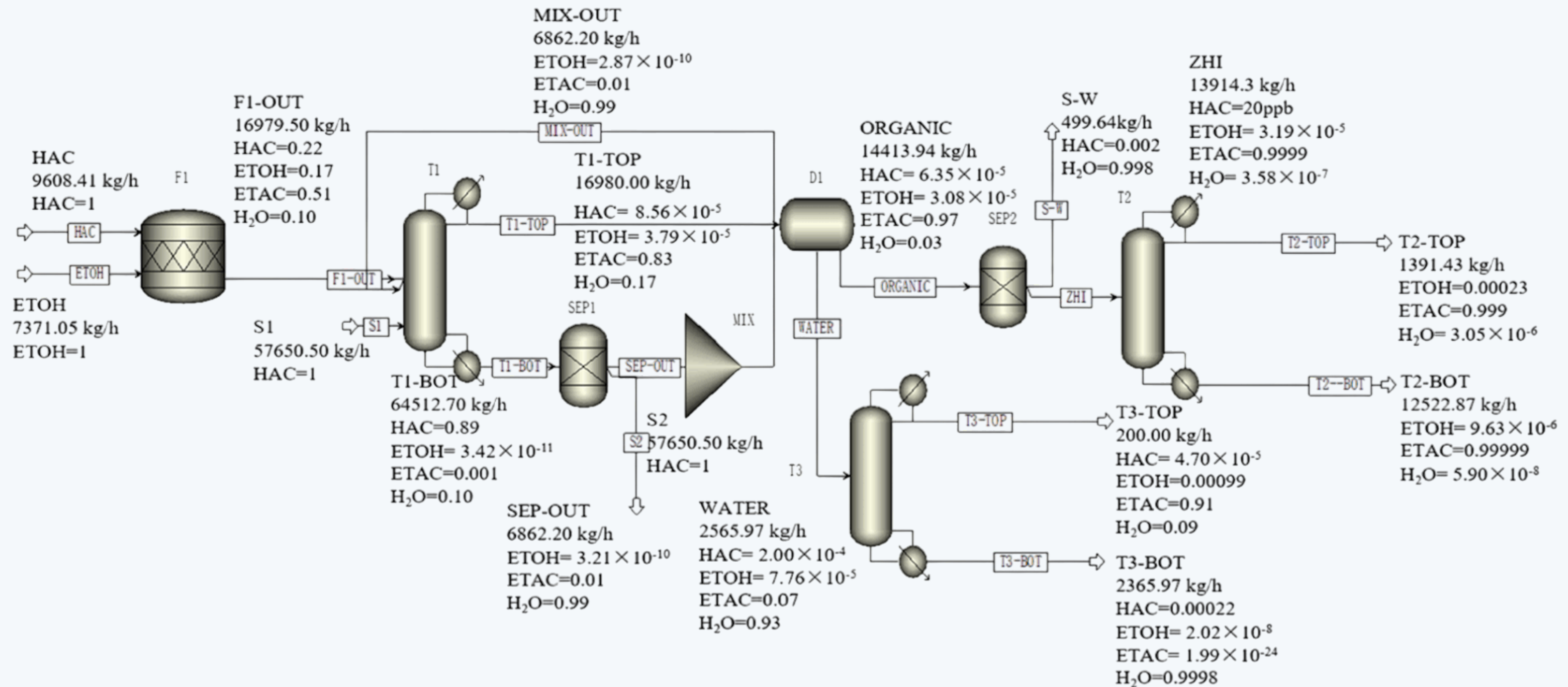
System	Tb (°C)	Composition
ETOH–ETAC–H ₂ O ⚡	70.50	0.14/0.78/0.08
ETAC–H ₂ O	70.99	0.88/0.12
ETOH–ETAC	71.81	0.30/0.70
ETAC (pure)	77.20	—
ETOH–H ₂ O	78.15	0.96/0.04
H ₂ O (pure)	100.02	—
HAC (pure)	118.01	—

Residue Curve Map — ETOH/ETAC/H₂O Ternary System



★ Operating in Region I ensures maximum ETAC at overhead

Reference Industrial Flowsheet (Xiao et al., 2025)



F1

Pre-Reactor

HAC + ETOH (1:1)
@ 70°C equilibrium

T1

Reactive Distillation

15× excess HAC
in reboiler (kettle)

D1

Phase Separator

Organic (96.5% ETAC)
vs Aqueous (92.9% H₂O)

SEP2

Mol. Sieve Membrane

Water → ppm levels
from organic phase

T2

Ester Purification

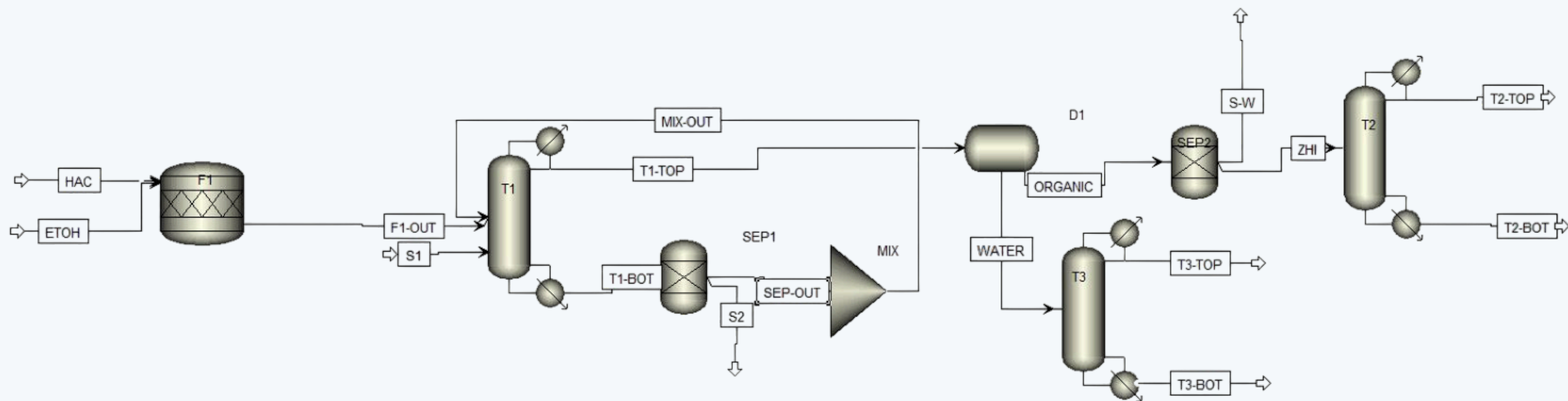
99.999 wt% ETAC
at column bottom

T3

Water Treatment

99.9 wt% H₂O
at column bottom

Group 2 — Aspen Plus Simulation Flowsheet



Key Stream Data (from Optimized Simulation)

Stream	Temp (°C)	Mass Flow (kg/hr)	ETAC (wt%)	H ₂ O (wt%)	HAC (wt%)
ETOH Feed	25.0	7,371	0.0	0.0	0.0
HAC Feed	25.0	9,608	0.0	0.0	100.0
T1-TOP	71.0	16,980	73.7%	17.0%	~0.0
ORGANIC (D1)	25.0	14,367	96.8%	3.2%	~0.0
T2-BOT (Product)	77.2	12,508	99.999%	~0.0	~0.0
T3-BOT (Water)	97.7	2,413	~0.0	99.8%	~0.0

Reaction Kinetics & Key Design Principles

Esterification Kinetics (Experimental — Solid Acid Catalyst)



$$r = 28,566.78 \cdot \exp(-40.15 / RT) \cdot C_{\text{ETOH}}$$

r = reaction rate ($\text{mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$) $E_a = 40.15 \text{ kJ/mol}$ T = temperature (K) *HAC:ETOH = 19:1 molar ratio*

Key Kinetic Parameters

Pre-exponential factor	28,566.78
Activation energy (E_a)	40.15 kJ/mol
Order in ETOH	1st order
Catalyst type	Solid acid resin (Amberlyst-36)
HAC:ETOH ratio	19:1 (excess HAC)
Reboiler temp range	~110°C

Key Design Principles

①

Reboiler-Only Reaction

All catalytic reaction confined to reboiler kettle — simplifies column internals & catalyst management. No reactive trays required.

②

Le Chatelier Driving Force

15× molar excess HAC pushes equilibrium toward complete ETOH conversion. ETOH eliminated from overhead entirely.

③

Binary Overhead Strategy

Without ETOH, overhead shifts to ETAC–H₂O binary azeotrope ($T_b=70.99^\circ\text{C}$) that cleanly stratifies via liquid-liquid separation.

④

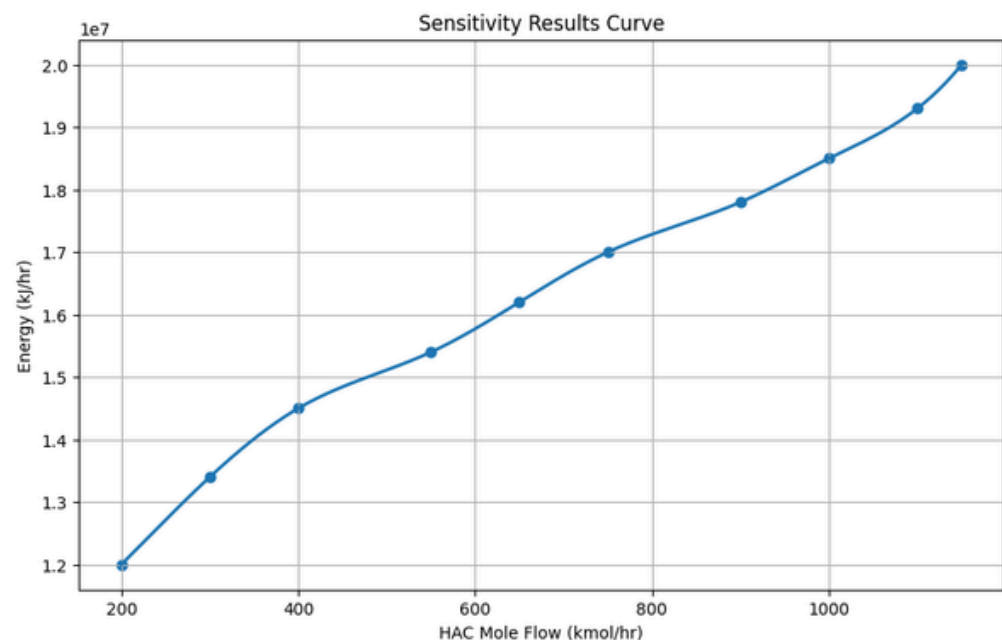
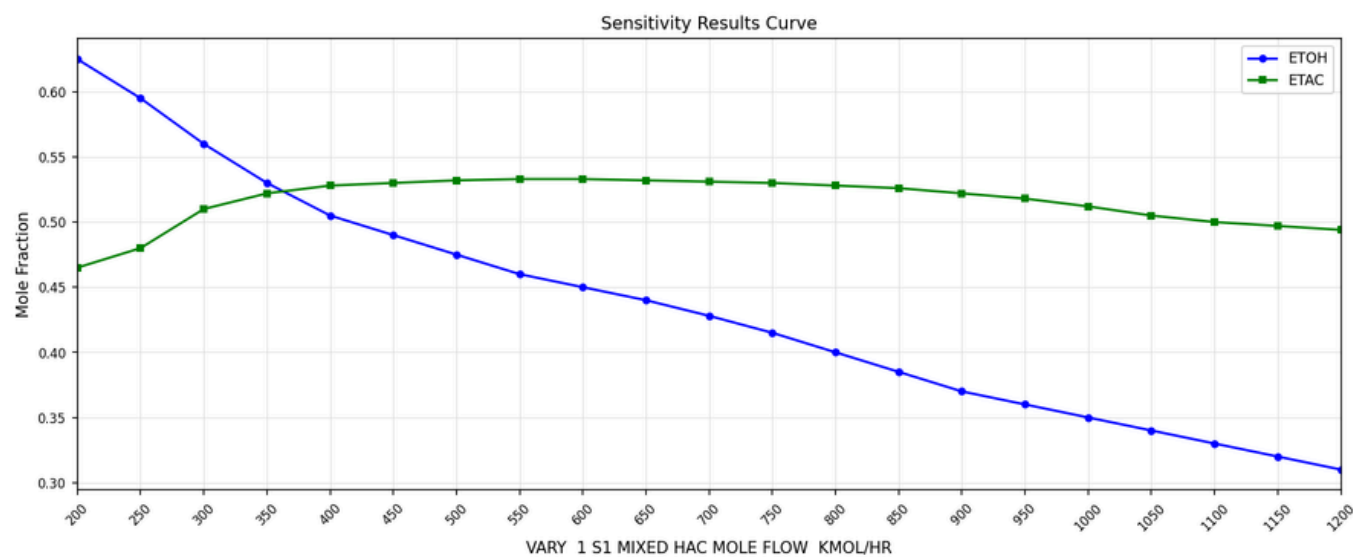
Low Reflux Enabled

Complete ETOH suppression removes need for high reflux — $RR=0.5$ sufficient, dramatically cutting reboiler duty vs. conventional RD.

Sensitivity Analysis — Column T1 (Reactive Distillation)

5.1 Effect of HAC Feed Rate

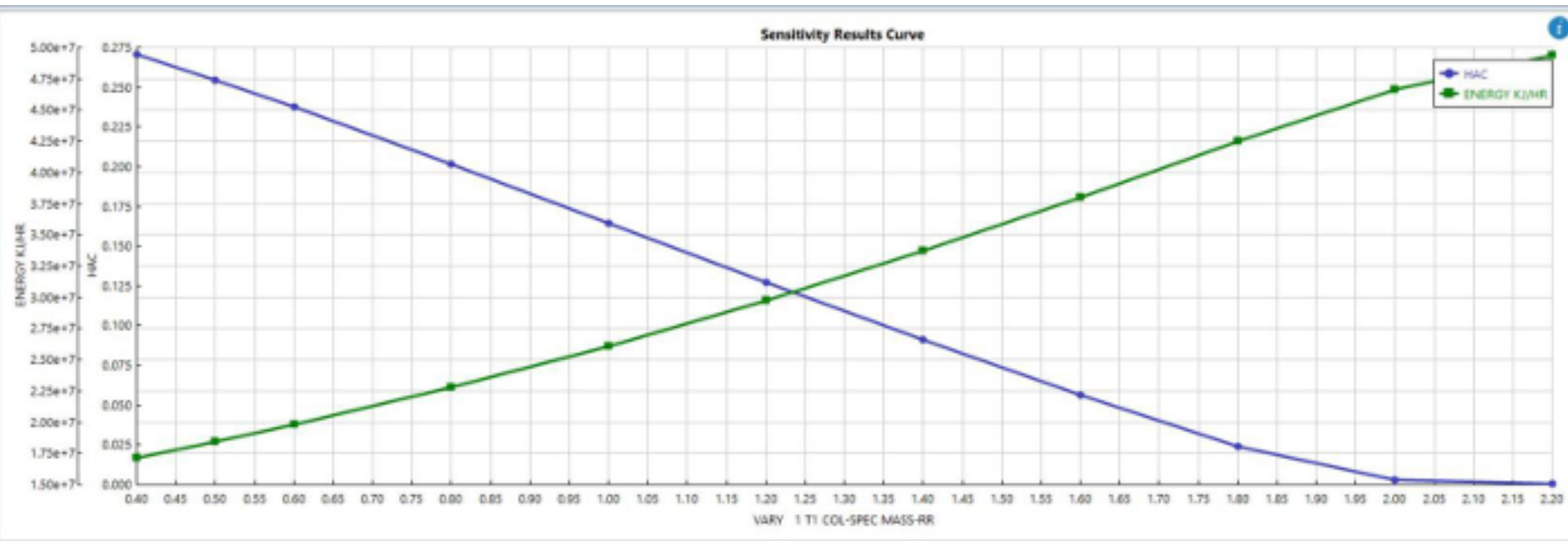
✓ Optimal: 960 kmol/hr



As HAC feed increases (200→1,200 kmol/hr), ETAC in distillate rises sharply and plateaus (~0.53 wt). ETOH→0. Energy rises monotonically (1.19→2.0 ×10⁷ kJ/hr). Trade-off: purity vs. energy.

5.3 Effect of Reflux Ratio

✓ Optimal: RR = 0.5

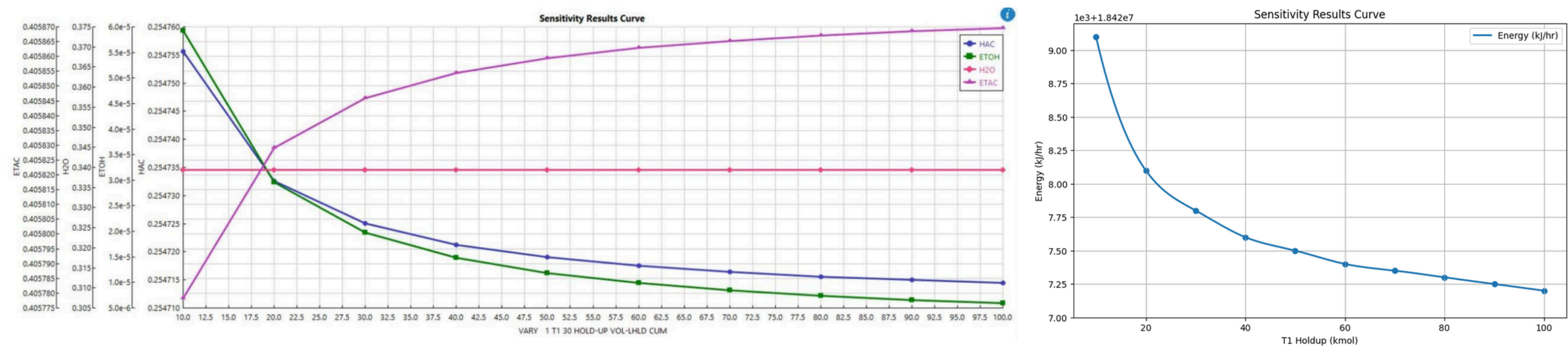


RR 0.40→2.20: HAC in distillate drops linearly (0.275→~0). Energy rises sharply (1.73→4.95 ×10⁷ kJ/hr). RR=0.5 gives acceptable HAC suppression at minimum energy — lowest viable ratio.

Sensitivity Analysis — Column T1 (Reactive Distillation)

5.2 Effect of Reboiler Holdup

✓ Optimal: 100 m³

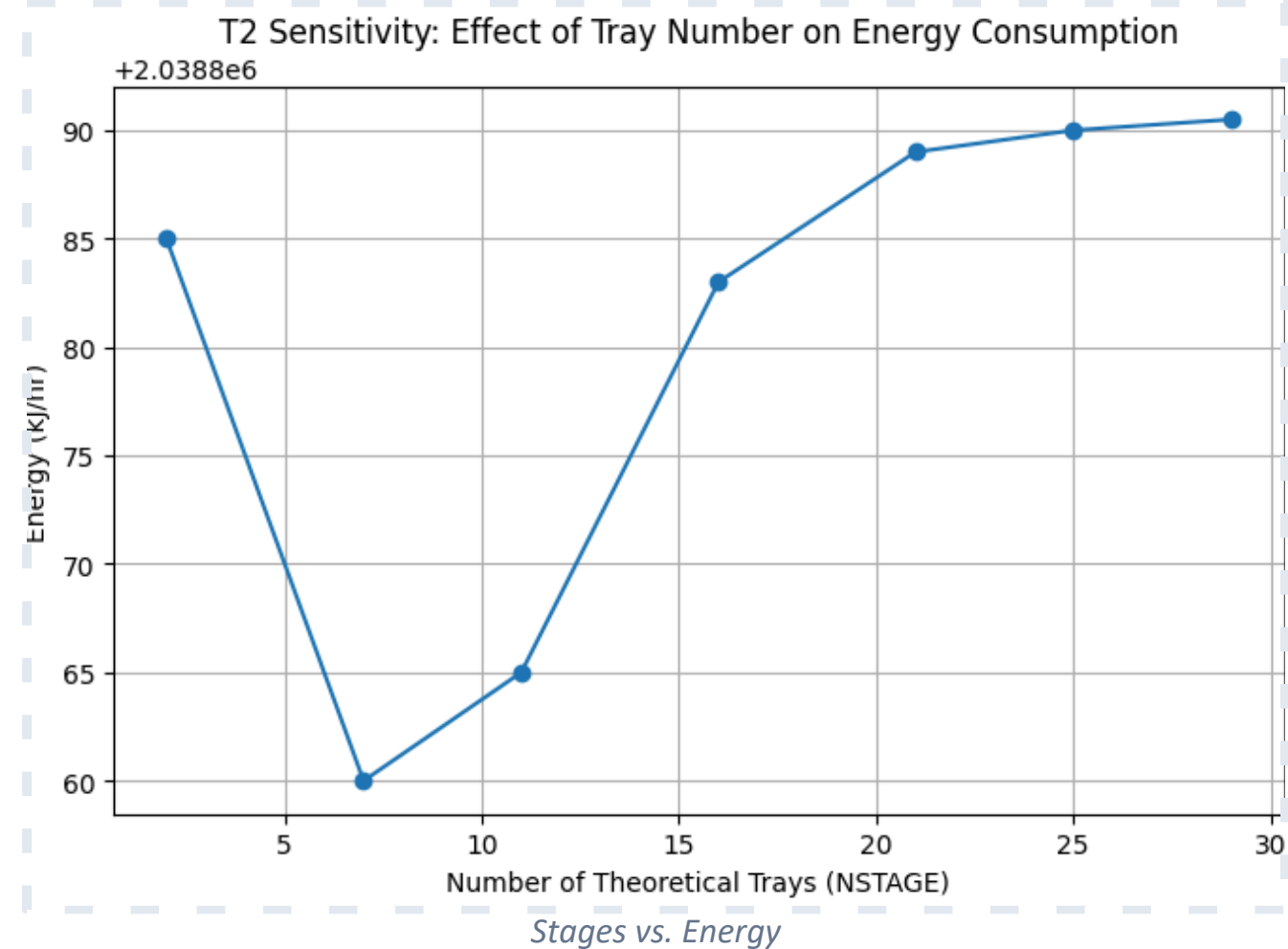


Holdup 10→100 m³ progressively lowers HAC & ETOH in overhead (HAC: $2.5 \times 10^{-4} \rightarrow 1.0 \times 10^{-4}$ wt). ETAC remains ~0.405. Energy slightly decreases — more residence time = more thorough reaction.

Sensitivity Analysis — Column T2 (Ester Purification) & T3 (Water Treatment)

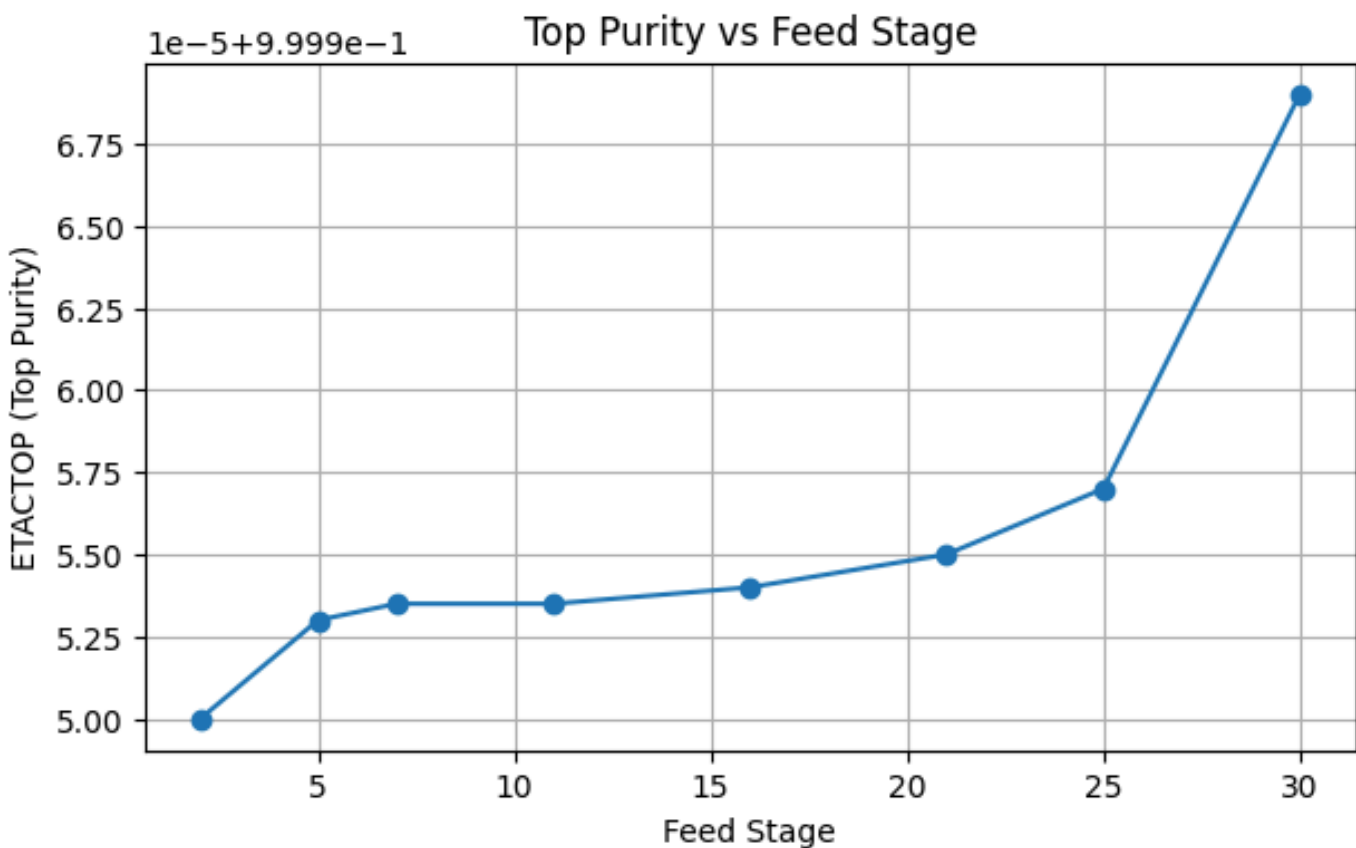
Column T2 — Ester Purification Column

No. of Stages



Optimal: 30 trays

Feed Stage



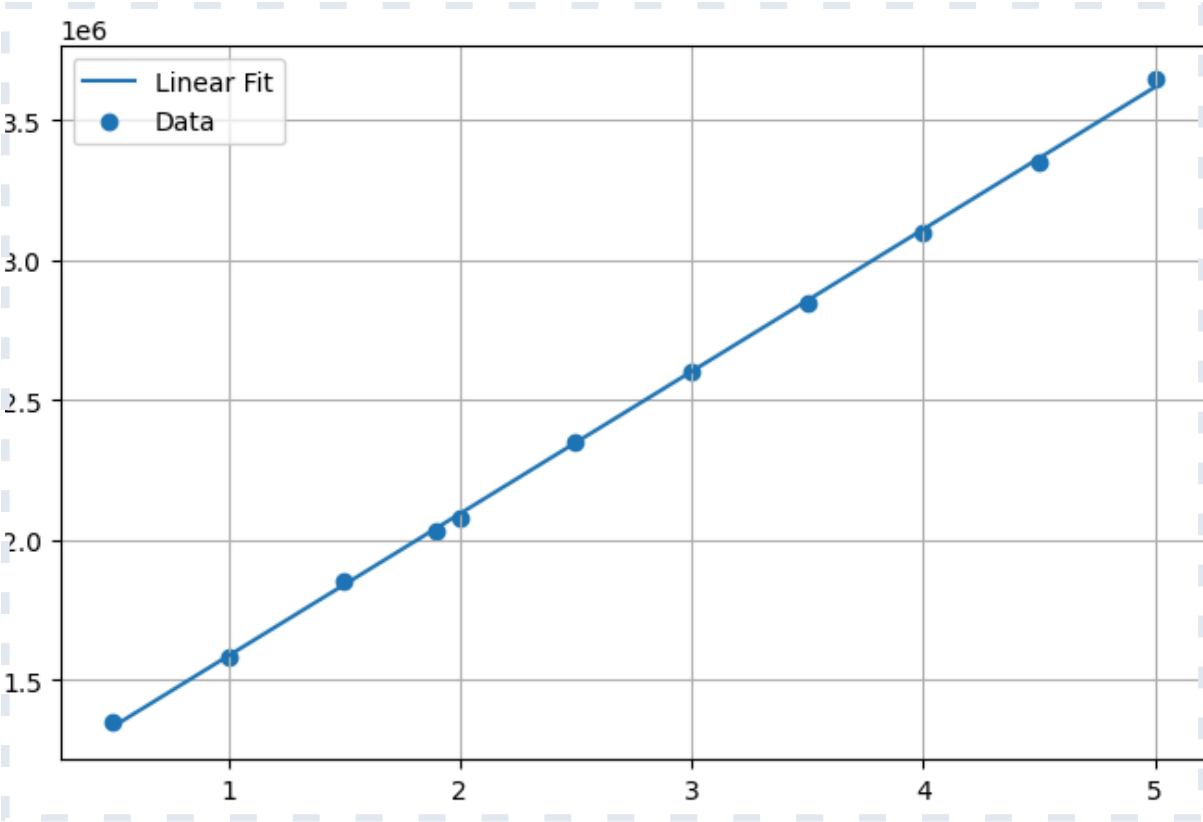
Optimal: 5th tray

Energy shows minimum at ~7 stages. At 30 stages ETAC purity in kettle = 99.999 wt%. Above 10 stages energy stabilizes — optimal balance at 30.

Purity peaks at 5th tray (99.999 wt%). Feed stage between 5–15 all meet spec. Energy minimized at feed stage 30 but purity compromise — 5th chosen.

Sensitivity Analysis — Column T2 (Ester Purification) & T3 (Water Treatment)

Reflux Ratio



RR vs. Energy

Optimal: RR = 1.9

Energy rises linearly $1.3 \times 10^6 \rightarrow 3.7 \times 10^6$ kJ/hr (RR 0.5 $\rightarrow$ 5.0). At $RR \geq 1.9$, bottom ETAC purity = 99.999 wt%. Minimum RR satisfying spec selected.

Column T3 — Water Treatment

Theoretical Trays $\rightarrow$ 20

Water purity stabilizes > 99.9% wt at 20+ trays. Investment cost balanced.

Feed Stage $\rightarrow$ 5th tray

Feed at 5th tray maximizes water purity at bottom. Energy also minimized here.

Reflux Ratio $\rightarrow$ RR = 0.7

Water purity stabilizes at $RR \geq 0.7$. Beyond this, energy increases without purity gain.

Top Draw Rate $\rightarrow$ 200 kg/hr

ETAC-rich top stream recycled. Water bottom: 99.9 wt% specification met.

Optimized Process Parameters — Summary

Parameter	T1 Reactive Distillation	T2 Ester Purification	T3 Water Treatment
Theoretical Trays	30	30	20
Reflux Ratio (mass)	0.5	1.9	0.7
Feed Location	Reboiler (kettle)	5th tray	5th tray
Top Withdrawal (kg/hr)	16,980	1,391.43	200.0
Reboiler Holdup (m³)	100	—	—
Additional HAC (kmol/hr)	960	—	—
Product Purity	83% ETAC (top)	99.999% ETAC (bottom)	99.9% H₂O (bottom)

1,149 kJ/kg

Total Energy Consumption

~80%

Energy Reduction vs. Conventional

99.999 wt%

Final ETAC Product Purity

99.9 wt%

Recovered Water Purity (T3)

Heat Integration Analysis — Aspen Energy Analyzer (AEA)

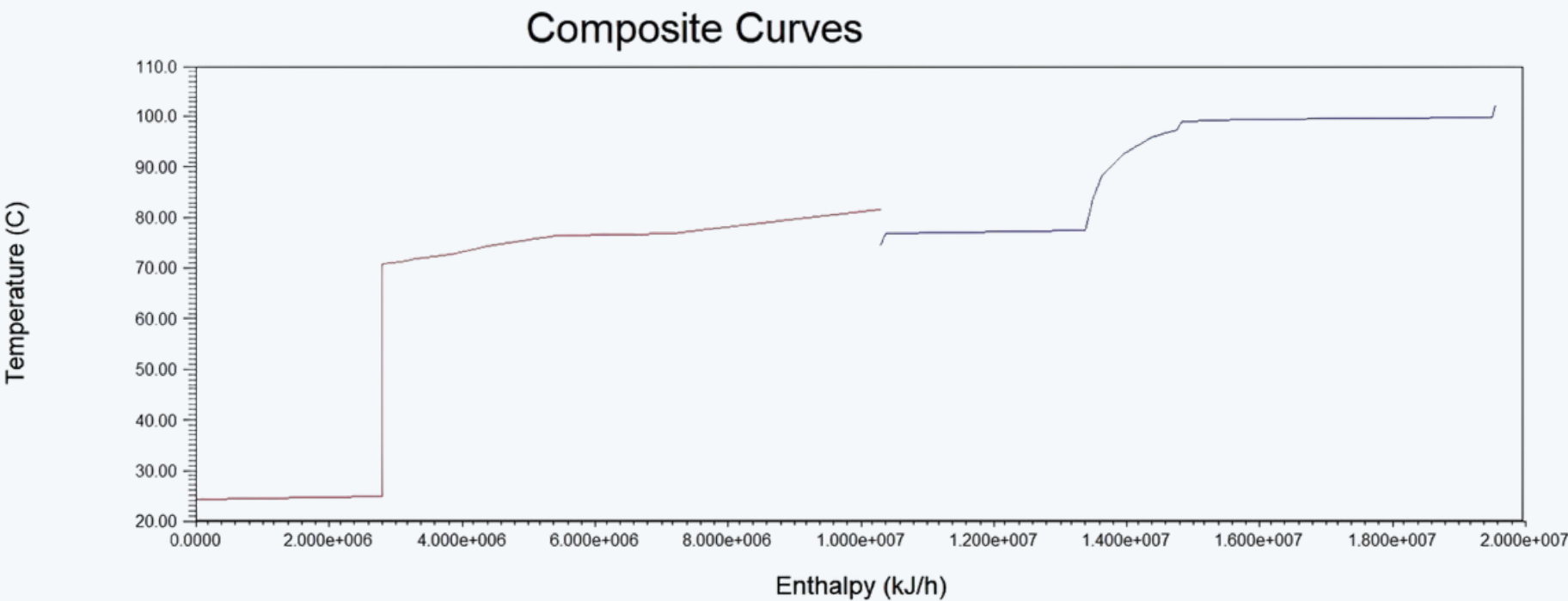
AEA Utility Summary ($\Delta T_{\min} = 10^{\circ}\text{C}$)

Utility Type	Actual (Gcal/hr)	Target (Gcal/hr)	Savings
Total Utilities	4.668	4.668	0.00 (0%)
Heating Utilities	2.201	2.201	0.00 (0%)
Cooling Utilities	2.467	2.467	0.00 (0%)
Carbon Emissions	0 kg/hr	0 kg/hr	0.00

Key Finding

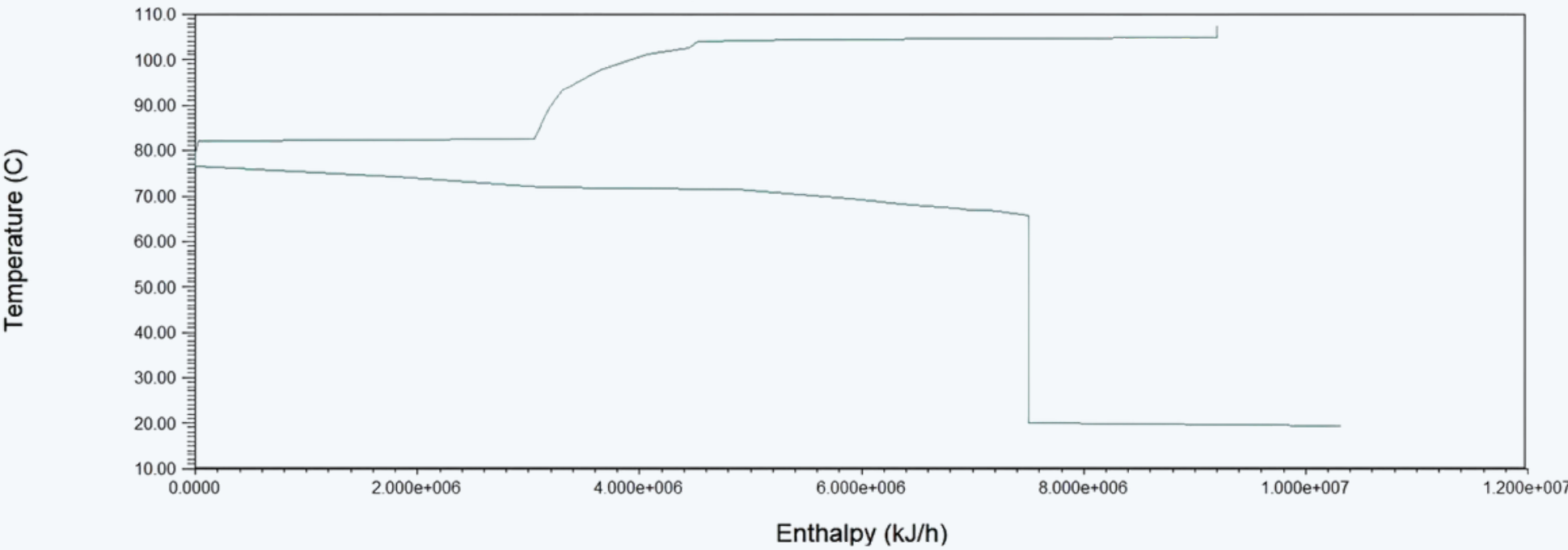
Zero available heat integration savings - the process is ALREADY operating at thermodynamic minimum energy. Actual = Target across ALL utilities. Carbon emissions = 0 (process-to-process heat exchange only, no fired heaters). Low-pressure steam (<1.2 bar) is the sole heating utility - possible because no utility is required above 105°C.

Hot & Cold Composite Curves



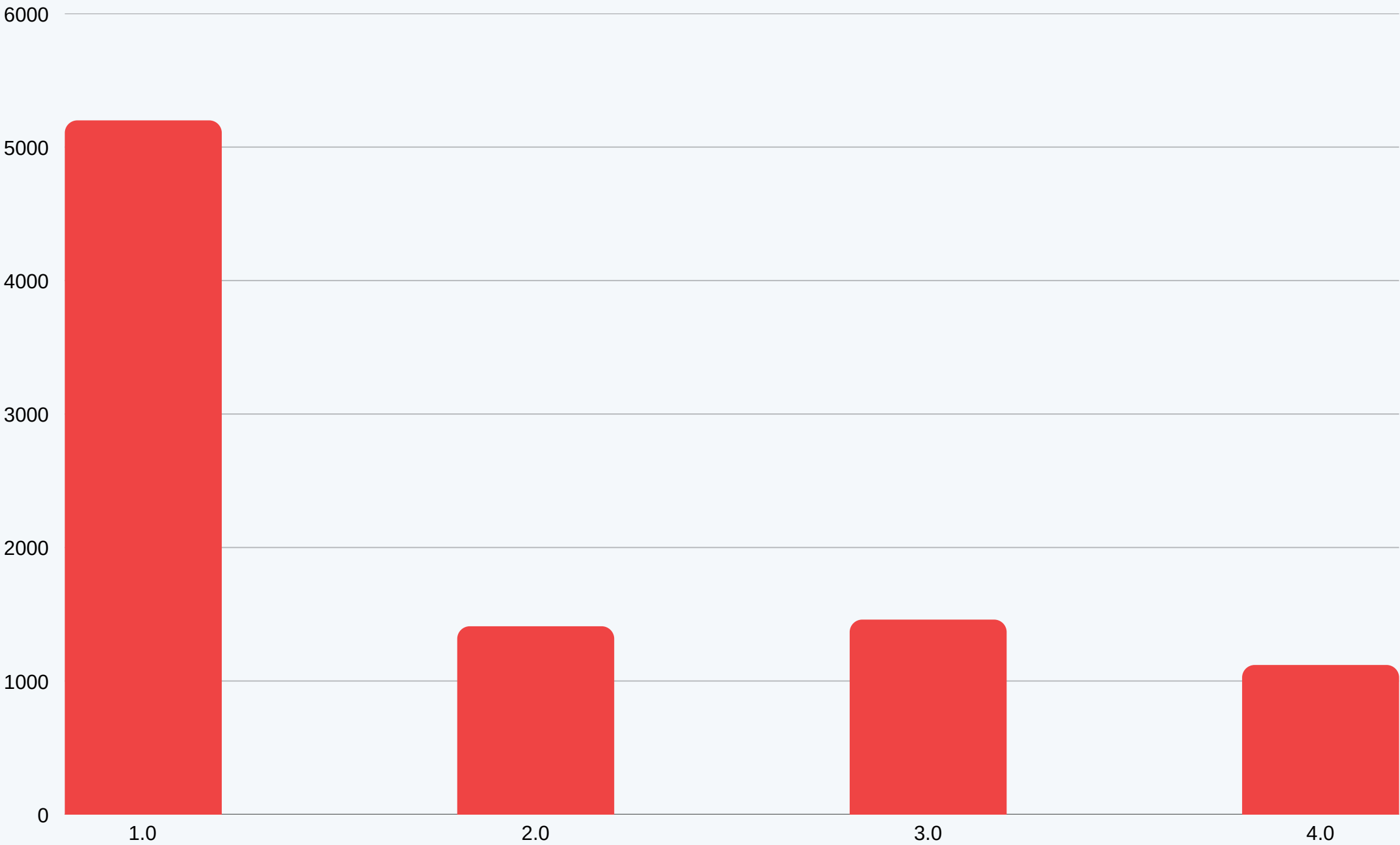
Tight pinch @ 75–80°C | Min. heat recovery possible | Plateaus = azeotrope phase changes

Grand Composite Curve (GCC)



Pockets at 75°C & 80°C | No utility needed above 105°C

Process Comparison — Literature & Industrial Benchmarking



Process Comparison Table

Process	Purity (wt%)	Energy (kJ/kg)
Traditional (Excess EtOH)	>99.5%	5,200+
Conventional RD	99.95%	1,410
ETAC-nBuAC Co-prod.	99.90%	1,460
This Work	99.999%	1,149

vs. Traditional Process

−80%

Energy saving

vs. Conventional RD

−21%

Energy saving

vs. Co-production

−23%

Energy saving

Purity Achievement

99.999%

Highest of all compared

Our Modification to the Reference Flowsheet

Introduction of Pinch Analysis (AEA) + Verification of Zero-Surplus Energy Configuration

The reference process (Xiao et al.) optimized T1/T2/T3 individually, but did not quantify whether further heat integration savings were available system-wide. We added an Aspen Energy Analyzer (AEA) pinch analysis step to validate the thermodynamic optimality of the entire flowsheet.

REFERENCE (Xiao et al., 2025)	OUR CONTRIBUTION (Group 2)
✗ Column-by-column sensitivity analysis only ✗ No system-wide heat integration check performed ✗ Energy savings from individual column optimization ✗ Total energy: 1,120 kJ/kg reported without HEN analysis ✗ No verification of thermodynamic limits	✓ Full AEA pinch analysis conducted ($\Delta T_{min} = 10^{\circ}C$) ✓ Composite curves & GCC generated and interpreted ✓ Confirmed actual = target utilities (ZERO surplus savings) ✓ Validated the process is at thermodynamic minimum ✓ GCC confirms low-pressure steam (<1.2 bar) as sole utility

Techno-Economic Analysis

Scenario B — Industry Benchmark | 100,000 TPA ETAC

Total Capital Inv.

\$38.5M

TCC + Working Capital

Annual Revenue

\$90.7M

@ \$0.90/kg ETAC

Net Profit (Yr 3)

\$9.14M

PAT after 25% tax

EBITDA Margin

15.4%

Yr 3 operations

Investment & Profitability Metrics

Simple Payback Period 4.1 years

Discounted Payback 5.9 years

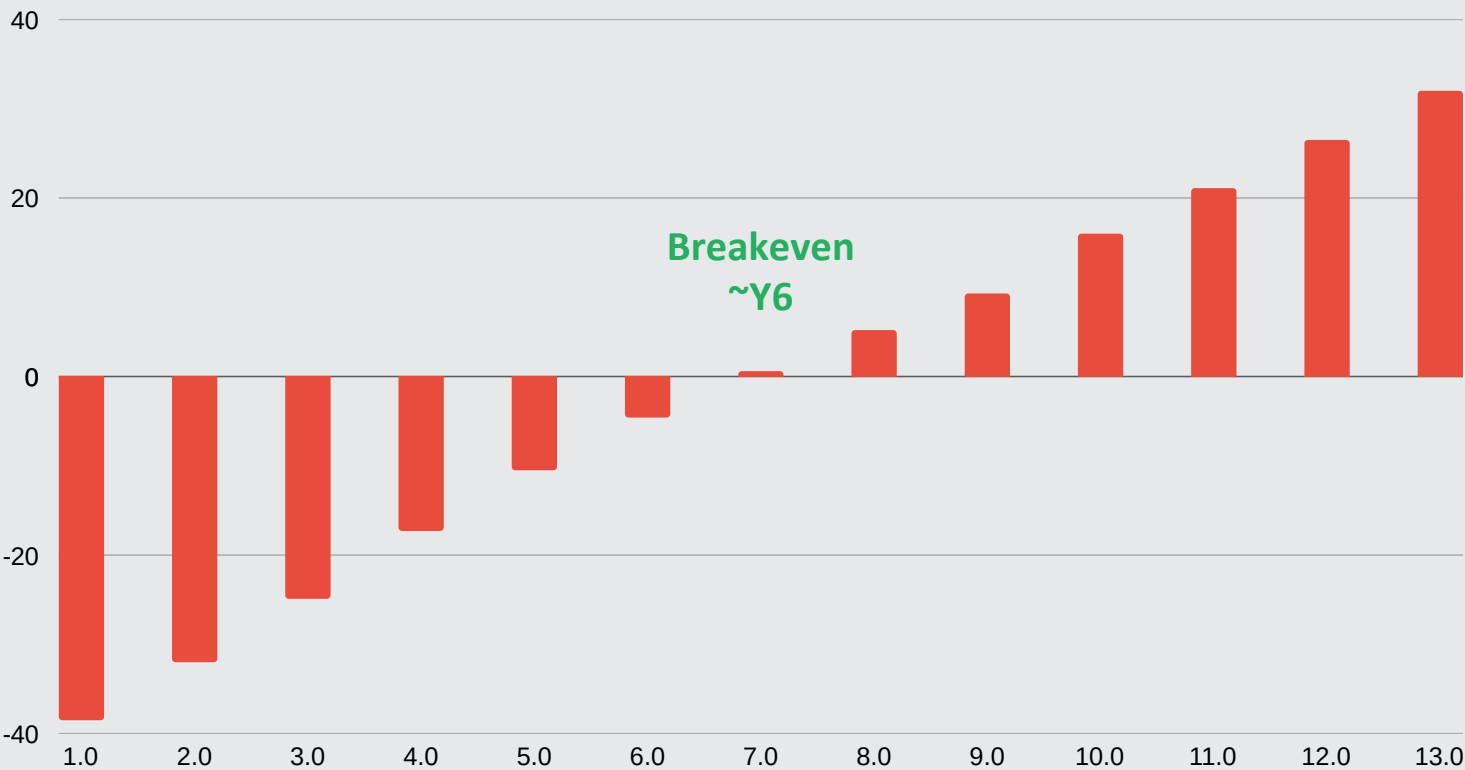
IRR (Internal Rate of Return) 24.1%

NPV @ 12% WACC (20 yr) \$32.0M

ROCE (Yr 3) 34.8%

Net Profit Margin (Yr 3) 10.1%

20-Year Cumulative NPV (Scenario B, \$M)



ETOH Feed: \$38.3M/yr

HAC Feed: \$34.6M/yr

Utilities: \$1.60M/yr

Fixed OpEx: \$2.24M/yr

Total OPEX: \$76.8M/yr

IRR 24.1% > WACC 12% ✓

Conclusions

1

A 15-fold molar excess of acetic acid in T1 reboiler achieves near-complete ethanol conversion, shifting the overhead to the binary ETAC–H₂O azeotrope — enabling direct phase stratification.

2

Optimal T1 parameters: 30 trays, holdup = 100 m³, reflux ratio = 0.5, HAC feed = 960 kmol/hr. Overhead product: 96.53 wt% ETAC — directly phase-separable.

3

Molecular sieve membrane (SEP2) reduces water in organic phase from 3.46% → ppm levels — a highly energy-efficient alternative to extractive distillation.

4

T2 achieves 99.999 wt% ETAC product (30 trays, feed @ 5th tray, RR=1.9). T3 recovers 99.9 wt% water (20 trays, feed @ 5th tray, RR=0.7).

5

Total energy consumption = 1,149 kJ/kg — representing an 80% reduction over traditional excess-ethanol processes and 21–23% improvement over other reactive distillation alternatives.

6

AEA pinch analysis confirms ZERO available heat integration savings — the process is already thermodynamically optimized, operating at the thermodynamic energy minimum.

References

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- [2] Santaella, M. A., Orjuela, A., & Narvaez, P. C. (2015). Comparison of different reactive distillation schemes for ethyl acetate production using sustainability indicators. Chemical Engineering and Processing, 96, 1–13.
- [3] Wu, Y. C., Lee, H. Y., Tsai, C. Y., Huang, H. P., & Chien, I. L. (2013). Design and control of a reactive-distillation process for esterification of an alcohol mixture containing ethanol and n-butanol. Computers and Chemical Engineering, 57, 63–77.
- [4] Smith, R. (2005). Chemical Process Design and Integration. Wiley.
- [5] Aspen Technology, Inc. (2023). Aspen Plus V14 User Guide. Bedford, MA.

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Thankyou